

# The empirical recurrence for OEIS A210104, proved by counting patterns

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## Abstract

OEIS A210104 counts arrays subject to a local condition *and* to the clause “new values 0..2 introduced in row major order”, which says the array is written in canonical form. That clause is not local, so a transfer matrix over the alphabet does not see it. It does not have to: what is being counted is not arrays but equality patterns, and the number of patterns is recovered from the numbers  $L_i$  of arrays over an  $i$ -letter alphabet by inverting a falling factorial. The result is the clean identity  $3!a(n) = \sum_i \binom{3}{i} D_{3-i} L_i(n)$  with  $D_m$  the derangement numbers, and each  $L_i$  is an ordinary walk count. A second reduction — the chain is strongly lumpable over the patterns, because the condition is relabelling-invariant — brings the state space down from 794 to  $S = 155$ . The entry’s empirical recurrence of order 58, contributed by R. H. Hardin, then follows from a finite exact computation.

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## 1 The sequence and the conjecture

OEIS A210104 is “Number of  $(n+1) \times 6$  0..2 arrays with every  $2 \times 2$  subblock having one or three distinct values, and new values 0..2 introduced in row major order.”. It has offset 1 and begins

2799, 95830, 3125615, 102841182, 3382058727, 111283430198, 3662188386511, 120530162560270, ...

The entry carries the following comment:

Empirical:  $a(n) = 68*a(n-1) - 1109*a(n-2) - 15132*a(n-3) + 533650*a(n-4) - 682812*a(n-5) - 91388848*a(n-6) + 541468084*a(n-7) + 8010220974*a(n-8) - 76037057944*a(n-9) - 380609964150*a(n-10) + 5713624684772*a(n-11) + 7546313419928*a(n-12) - 272240689140684*a(n-13) + 185952132990252*a(n-14) + 8790130916081940*a(n-15) - 18031849778390353*a(n-16) - 197980986867193344*a(n-17) + 636160628144662267*a(n-18) + 3122695677229408840*a(n-19) - 14015802619871969522*a(n-20) - 33484157496106096408*a(n-21) + 214830105901029340796*a(n-22) + 215490176857593683592*a(n-23) - 2386196515012527746772*a(n-24) - 278928178726785919580*a(n-25) + 19547830951977668052136*a(n-26) - 10058031406426519679456*a(n-27) - 118783413433906506268928*a(n-28) + 120366274979665650042448*a(n-29) + 533880171597066533206896*a(n-30) - 778900186987919254121344*a(n-31) - 1754277169343190873917504*a(n-32) + 3380528371752993713671808*a(n-33) + 4110607686089791274598592*a(n-34) - 10417505934114336408844032*a(n-35) - 6508643137572987893783936*a(n-36) + 23248657105901998508654080*a(n-37) + 5969723267297107040175616*a(n-38) - 37777911031496989203732480*a(n-39) - 751674288908450157042688*a(n-40) + 44600619081046898666690560*a(n-41) - 5794844778782401587208192*a(n-42) - 37966938282556065993916416*a(n-43) + 8160526737847235042664448*a(n-44) + 23005895699982159363768320*a(n-45) - 5661409259903212447334400*a(n-46) - 9727195323636513269350400*a(n-47) + 2227995919449608334737408*a(n-48) + 2777538507080041554247680*a(n-49) - 472156158151071247630336*a(n-50) - 504235867339370029645824*a(n-51) + 41294805044945664081920*a(n-52) + 51365990278841539493888*a(n-53) + 588902075096209817600*a(n-54) - 2181425252011962531840*a(n-55) - 163298906545663770624*a(n-56) + 18362493445475401728*a(n-57) + 1626733500608544768*a(n-58)$

As of the “Last modified” line on the live entry (Oct 03 2025 15:43:34, revision 11) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 68a(n-1) - 1109a(n-2) - 15132a(n-3) + 533650a(n-4) - 682812a(n-5) - 91388848a(n-6) + 54146808 \quad (1)$$

for all  $n > 57$ .

## 2 What the canonical-form clause counts

Fix the shape: 6 columns and  $L = n + 1$  rows, so  $L \cdot 6$  cells. Call a partition of the cells into nonempty classes a *pattern*; an array over an alphabet determines a pattern, two cells lying in the same class exactly when they carry the same value.

Every  $2 \times 2$  subblock of the array occupies two consecutive rows and two consecutive columns; writing those two rows as  $r$  and  $s$  the subblock is

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r_j & r_{j+1} \\ s_j & s_{j+1} \end{pmatrix},$$

and the entry’s requirement on it is

$$\#\{\alpha, \beta, \gamma, \delta\} \in \{1, 3\}. \quad (2)$$

The condition is stated purely in terms of equalities between entries, so it is invariant under any relabelling of the values and is therefore a property of the pattern alone. Call a pattern *admissible* when it has the property.

The clause “new values 0..2 introduced in row major order” selects, from each class of arrays that differ only by a relabelling, exactly one representative: reading the cells in row major order, the first occurrence of the value  $v$  must precede the first occurrence of  $v + 1$ . Every pattern with at most  $K = 3$  classes has exactly one such representative, and every array in canonical form is the representative of its own pattern. Hence, writing  $N_j(n)$  for the number of admissible patterns with exactly  $j$  classes,

$$a(n) = \sum_{j=0}^3 N_j(n). \quad (3)$$

**Lemma 1.** *Let  $L_i(n)$  be the number of arrays over an alphabet of  $i$  letters satisfying the local condition, with no canonical-form clause. Then*

$$L_i(n) = \sum_j N_j(n) i(i-1) \cdots (i-j+1), \quad 3! \sum_{j=0}^3 N_j(n) = \sum_{i=0}^3 \binom{3}{i} D_{3-i} L_i(n),$$

where  $D_m = m! \sum_{t=0}^m (-1)^t / t!$  is the number of derangements of  $m$  letters.

*Proof.* An array over  $i$  letters satisfying the condition determines an admissible pattern together with an injection of its classes into the alphabet, and conversely; a pattern with  $j$  classes admits  $i(i-1) \cdots (i-j+1)$  injections. That is the first identity. Writing  $i(i-1) \cdots (i-j+1) = j! \binom{i}{j}$  it reads  $L_i = \sum_j (j! N_j) \binom{i}{j}$ , so by binomial inversion  $j! N_j = \sum_i (-1)^{j-i} \binom{j}{i} L_i$ . Summing over  $j \leq 3$  and exchanging the order of summation,

$$\sum_{j=0}^3 N_j = \sum_{i=0}^3 L_i \sum_{j=i}^3 \frac{(-1)^{j-i}}{i! (j-i)!} = \sum_{i=0}^3 \frac{L_i}{i!} \sum_{t=0}^{3-i} \frac{(-1)^t}{t!},$$

and multiplying by  $3!$  turns the coefficient of  $L_i$  into  $\frac{3!}{i!} \sum_{t \leq 3-i} (-1)^t / t! = \binom{3}{i} (3-i)! \sum_{t \leq 3-i} (-1)^t / t! = \binom{3}{i} D_{3-i}$ , an integer.  $\square$

### 3 Each $L_i$ is a walk count, and the chain lumps

Fix  $i$  and let  $V_i = \{0, \dots, i-1\}^6$  be the possible rows. For  $r, s \in V_i$  declare  $r \rightarrow s$  an edge when placing  $s$  directly after  $r$  satisfies (2) at every column, and let  $M_i$  be the adjacency matrix. An array is exactly a sequence of rows, and its condition is exactly the conjunction of the edge conditions along that sequence, so

$$L_i(n) = \mathbf{s}_i^\top M_i^n \mathbf{1}.$$

Because the condition involves no row before the first, every row may start a walk, and  $\mathbf{s}_i = \mathbf{1}$ .

**Lemma 2.** *Let  $\pi(r)$  be the equality pattern of the row  $r$ . For all  $r, r'$  with  $\pi(r) = \pi(r')$  and every pattern  $Q$ ,*

$$\#\{s : r \rightarrow s, \pi(s) = Q\} = \#\{s : r' \rightarrow s, \pi(s) = Q\}.$$

*Consequently the numbers  $\widehat{M}_i[P][Q]$  obtained by taking any representative of  $P$  are well defined, and with  $e_P = i(i-1) \cdots (i-|P|+1)$  the number of rows of pattern  $P$ ,*

$$\mathbf{s}_i^\top M_i^m \mathbf{1} = \sum_P \varepsilon_P f_m(P), \quad f_0 \equiv 1, \quad f_{m+1}(P) = \sum_Q \widehat{M}_i[P][Q] f_m(Q),$$

*where  $\varepsilon_P = e_P$  if rows of pattern  $P$  may start a walk and 0 otherwise.*

*Proof.* If  $\pi(r) = \pi(r')$  then the map sending the value of  $r$  at a column to the value of  $r'$  at that column is a well-defined injection between the letters used, and extends to a permutation  $\sigma$  of the alphabet with  $r' = \sigma(r)$ . The edge condition is a statement about equalities among entries, so  $r \rightarrow s$  if and only if  $\sigma(r) \rightarrow \sigma(s)$ ; and  $\pi(\sigma(s)) = \pi(s)$ . Hence  $s \mapsto \sigma(s)$  is a bijection between the two successor sets preserving patterns, which is the first claim. It says the chain is strongly lumpable over  $\pi$ , so  $(M_i^m \mathbf{1})(r)$  depends on  $r$  only through  $\pi(r)$ ; calling that value  $f_m(\pi(r))$  gives the recursion, and summing over  $r$  in each class gives the displayed identity, the number of rows of pattern  $P$  being the number of injections of its  $|P|$  classes into the alphabet.  $\square$

Assembling the 3 lumped chains into one block-diagonal matrix  $N$  of size  $S = 155$  and putting the weights of Lemma 1 into the start vector,  $\mathbf{v} = \left(\binom{3}{i} D_{3-i} \varepsilon_P\right)_{i,P}$ , gives

$$3! a(n) = \mathbf{v}^\top N^n \mathbf{1}. \tag{4}$$

The unlumped alphabet had 794 rows in all; the lumped chain has  $S = 155$  states.

### 4 The criterion, and the computation

Let  $q(t) = t^{58} - \sum_i c_i t^{58-i}$  be the characteristic polynomial of (1).

**Lemma 3.** *For every  $n$  with  $m = n + 1 - 1 \geq 58$ ,*

$$3! \left( a(n) - \sum_i c_i a(n-i) \right) = \mathbf{v}^\top N^{m-58} q(N) \mathbf{1}.$$

*Hence (1) holds from that point on if and only if  $\mathbf{v}^\top N^j q(N) \mathbf{1} = 0$  for every  $j \geq 0$ , and it is enough to check  $j = 0, 1, \dots, S-1$ .*

*Proof.* Substitute (4) into the left side and factor  $N^{m-58}$  out of  $N^m - \sum_i c_i N^{m-i}$ . The scalar  $3!$  is nonzero, so it does not affect whether the left side vanishes. For the last clause put  $w = q(N) \mathbf{1}$ : by Cayley–Hamilton  $N^S$  is an integer combination of  $I, N, \dots, N^{S-1}$ , so each  $\mathbf{v}^\top N^j w$  with  $j \geq S$  is the corresponding combination of earlier values.  $\square$

**Theorem 4.** *The recurrence (1) holds for every  $n > 57$ , which is the statement of Section 1.*

*Proof.* The lumped transition counts were obtained by enumerating, for one representative row of each pattern, all admissible successors and sorting them by pattern — the successors being generated column by column, since the edge condition is a conjunction of one constraint per column. The vector  $w = q(N)\mathbf{1}$  was then formed by 58 matrix–vector products in exact integer arithmetic and  $\mathbf{v}^\top N^j w$  evaluated for  $j = 0, 1, \dots$ , again exactly. Every one of those integers vanishes from the index corresponding to  $n = 57 + 1$  onwards, and the run continued until  $w$  was the zero vector, which settles all larger  $j$  at once. Lemma 3 gives the theorem.  $\square$

## 5 Verification

Three checks, each independent of the algebra above.

First, the whole construction was compared against the entry’s DATA: the right-hand side of (4), divided by  $3!$ , was evaluated at every one of the 14 published indices and agrees exactly. This is a strong test of the modelling and not only of the arithmetic — the inversion of Lemma 1, the reading of the local condition, and the alignment of the entry’s index with the number of rows would each show up as a mismatch, and the quotient by  $3!$  came out an integer at every index.

Second, the lumped chain was checked against the unlumped one: for the small shapes the walk counts  $\mathbf{s}_i^\top M_i^m \mathbf{1}$  were computed directly over the full alphabet as well and agree with  $\sum_P \varepsilon_P f_m(P)$  term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 3 was run until the working vector was identically zero rather than to a sampled prefix.

## References

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